

Department of Engineering Cybernetics
TTK4215 Adaptive Control
Solution 3

Problem 1 of Chapter 2 of Ioannou & Fidan

We have

$$(s^3 + a_2s^2 + a_1s + a_0)[y] = (b_2s^2 + b_1s + b_0)[u] \quad (1)$$

so defining $\Lambda(s) = s^3 + \lambda_2s^2 + \lambda_1s + \lambda_0$, we have

a)

$$\underbrace{\frac{s^3}{\Lambda(s)}[y]}_z = \underbrace{[\begin{matrix} a_2 & a_1 & a_0 & b_2 & b_1 & b_0 \end{matrix}]}_{\theta^{*T}} \underbrace{\begin{bmatrix} -\frac{s^2}{\Lambda(s)}[y] \\ -\frac{s}{\Lambda(s)}[y] \\ -\frac{1}{\Lambda(s)}[y] \\ \frac{s^2}{\Lambda(s)}[u] \\ \frac{s}{\Lambda(s)}[u] \\ \frac{1}{\Lambda(s)}[u] \end{bmatrix}}_{\phi},$$

b)

$$\underbrace{\frac{s^3 + 3s^2 + s + 2}{\Lambda(s)}[y]}_z = \underbrace{[\begin{matrix} b_2 & b_1 & b_0 \end{matrix}]}_{\theta^{*T}} \underbrace{\begin{bmatrix} \frac{s^2}{\Lambda(s)}[u] \\ \frac{s}{\Lambda(s)}[u] \\ \frac{1}{\Lambda(s)}[u] \end{bmatrix}}_{\phi},$$

and

c)

$$\underbrace{\frac{s^3}{\Lambda(s)}[y] - \frac{2}{\Lambda(s)}[u]}_z = \underbrace{\begin{bmatrix} a_2 & a_1 & a_0 \end{bmatrix}}_{\theta^{*T}} \underbrace{\begin{bmatrix} -\frac{s^2}{\Lambda(s)}[y] \\ -\frac{s}{\Lambda(s)}[y] \\ -\frac{1}{\Lambda(s)}[y] \end{bmatrix}}_{\phi}.$$

Problem 2 of Chapter 2 of Ioannou & Fidan

Move $\frac{100s^2}{\Lambda(s)}$ to the right hand side of (2.6) and redefine z, ϕ, θ^* accordingly.

Problem 3 of Chapter 2 of Ioannou & Fidan

Arranging known terms on the left hand side and uncertain ones on the right hand side, we get

$$\underbrace{y(k) + 1.3y(k-1) - u(k-2)}_z = \underbrace{\begin{bmatrix} a_2 & b_1 \end{bmatrix}}_{\theta^{*T}} \underbrace{\begin{bmatrix} -y(k-2) \\ u(k-1) \end{bmatrix}}_{\phi}. \quad (2)$$

Problem 5 of Chapter 2 of Ioannou & Fidan

a)

Let

$$z = \frac{s^2 + 2s + 1}{s^2 + \lambda_1 s + \lambda_0} [x] \quad (3)$$

and

$$\begin{aligned} \theta^* &= \begin{bmatrix} a_1 & a_2 & b_1 & b_2 \end{bmatrix}^T \\ \phi &= \begin{bmatrix} \frac{1}{s^2 + \lambda_1 s + \lambda_0} [f_1(x)] & \frac{1}{s^2 + \lambda_1 s + \lambda_0} [f_2(x)] & \frac{1}{s^2 + \lambda_1 s + \lambda_0} [g_1(x)u] & \frac{1}{s^2 + \lambda_1 s + \lambda_0} [g_2(x)u] \end{bmatrix}^T \end{aligned} \quad (4)$$

and we have the SPM

$$z = \theta^{*T} \phi. \quad (6)$$

b)

Let

$$z = \frac{s^2 + 2s + 1}{s^2 + \lambda_1 s + \lambda_0} [x] \quad (7)$$

and

$$\theta^* = [a_1 \ a_2 \ b_1 \ b_2]^T \quad (8)$$

$$W(s) = \frac{1}{s^2 + \lambda_1 s + \lambda_0} \quad (9)$$

$$\phi = [f_1(x) \ f_2(x) \ g_1(x)u \ g_2(x)u]^T \quad (10)$$

and we have the DPM

$$z = W(s)[\theta^{*T} \phi]. \quad (11)$$

Problem 6 of Chapter 2 of Ioannou & Fidan

We have

$$(s^2 + as + c)[y] = K_p(s + b)[u]$$

so

$$s^2 y = K_p s u + K_p b u - a s y - c y.$$

Defining $\Lambda(s) = s^2 + \lambda_1 s + \lambda_0$ (must be Hurwitz) and $z = s^2/\Lambda(s)[y]$ we have

$$z = K_p \frac{s}{\Lambda(s)}[u] + K_p b \frac{1}{\Lambda(s)}u - a \frac{s}{\Lambda(s)}[y] - c \frac{1}{\Lambda(s)}[y].$$

a)

Define $\rho^* = K_p$, $\theta^* = [b \ a/K_p \ c/K_p]^T$, $\phi = [1/\Lambda(s)[u] \ -s/\Lambda(s)[y] \ -1/\Lambda(s)[y]]^T$ and $z_1 = s/\Lambda(s)[u]$ to get the B-SPM (2.3).

b)

Define $P(s) = s + p_0$ for a $p_0 > 0$, $W(s) = P(s)/\Lambda(s)$, $\rho^* = K_p$, $\theta^* = [b \ a/K_p \ c/K_p]^T$, $\phi = [1/P(s)[u] \ -s/P(s)[y] \ -1/P(s)[y]]^T$ and $z_1 = s/P(s)[u]$ to get the B-DPM (2.4).

c)

Define $\theta^* = [K_p \ K_p b \ a \ c]^T$ and $\phi = [s/\Lambda(s)[u] \ 1/\Lambda(s)[u] \ -s/\Lambda(s)[y] \ -1/\Lambda(s)[y]]^T$ to get the linear SPM (2.1).

d)

Define $P(s) = s + p_0$ for a $p_0 > 0$, $W(s) = P(s)/\Lambda(s)$, $\theta^* = [K_p \ K_p b \ a \ c]^T$ and $\phi = [s/P(s)[u] \ 1/P(s)[u] \ -s/P(s)[y] \ -1/P(s)[y]]^T$ to get the linear DPM (2.2).

Problem 7 of Chapter 2 of Ioannou & Fidan

We have

$$\begin{aligned} \dot{x} &= \sum_i^m W_{fi}^* \varphi_{fi}(x) + \sum_i^n W_{gi}^* \varphi_{gi}(x)u \\ &= \underbrace{[W_{f1} \ \cdots \ W_{fm} \ W_{g1} \ \cdots \ W_{gn}]}_{\theta^{*T}} \underbrace{\begin{bmatrix} \varphi_{f1}(x) \\ \vdots \\ \varphi_{gm}(x) \\ \varphi_{g1}(x)u \\ \vdots \\ \varphi_{gn}(x)u \end{bmatrix}}_{\bar{\phi}}. \end{aligned} \quad (12)$$

So we have

$$\underbrace{\frac{s}{s + \lambda_0}[x]}_z = \theta^{*T} \underbrace{\frac{1}{s + \lambda_0}[\bar{\phi}]}_{\phi}. \quad (13)$$